

Homework 1, Math 609

August 31, 2026

PROBLEM 1: (10 points)

- (5 points) A monomial matrix is a square matrix in which each row and column contains exactly one nonzero entry. Prove that a monomial matrix is nonsingular.
- (5 points) A square matrix A is said to be skew-symmetric if $A^T = -A$. Prove that if A is skew-symmetric, then $x^T Ax = 0$ for all x .

PROBLEM 2: (10 points)

- (5 points) Given

$$E_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ \ell_{21} & 1 & 0 & 0 \\ \ell_{31} & 0 & 1 & 0 \\ \ell_{41} & 0 & 0 & 1 \end{pmatrix}, \quad P_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix},$$

find $\tilde{E}_1$ such that $P_2 E_1 = \tilde{E}_1 P_2$.

- (5 points) In general, let P_j^i denote the permutation matrix which interchanges row i and row j . Let E_k be an elementary row operation matrix used in Gaussian elimination. Show that if $i > j > k$, then $P_j^i E_k = \tilde{E}_k P_j^i$, where $\tilde{E}_k$ has the same form as E_k . How are E_k and $\tilde{E}_k$ different?

PROBLEM 3: (10 points)

- (5 points) Given

$$A = \begin{pmatrix} 6 & 10 & 0 \\ 12 & 26 & 4 \\ 0 & 9 & 12 \end{pmatrix}, \quad b = \begin{pmatrix} 4 \\ 10 \\ -3 \end{pmatrix},$$

find the LU factorization of A . Then solve the system $Ax = b$ by forward elimination ($Ly = b$) and back substitution ($Ux = y$).

- (5 points) Let

$$A = \begin{pmatrix} 1 & -5 & 1 \\ 10 & 0 & 20 \\ 5 & 0 & -1 \end{pmatrix}, \quad b = \begin{pmatrix} 7 \\ 6 \\ 4 \end{pmatrix}.$$

Find a permutation matrix P such that $PA = LU$. Write U and L . Solve $PAx = Pb$ by forward elimination ($Ly = Pb$) and back substitution ($Ux = y$).

PROBLEM 4: (10 points) Consider the $N \times N$ tridiagonal matrix

$$A = \frac{1}{h^2} \begin{pmatrix} 2\varepsilon + h^2 & -\varepsilon & 0 & \dots & 0 & 0 & 0 \\ -\varepsilon & 2\varepsilon + h^2 & -\varepsilon & \dots & 0 & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & -\varepsilon & 2\varepsilon + h^2 & -\varepsilon \\ \dots & \dots & \dots & \dots & 0 & -\varepsilon & 2\varepsilon + h^2 \end{pmatrix}$$

and the vector $f = (f_1, \dots, f_N)^T$, where $f_i = 2ih + 1$, $h = 2^{-n}$, $N = 2^n - 1$, $n = 1, 2, \dots, 8$. Take $\varepsilon = 10^{-3}$ and solve the system $Au = f$ using LU factorization for tridiagonal matrix as discussed in the notes. In your code do not create the matrix A . For each value of $n = 1, 2, \dots, 8$, plot the points (x_i, u_i) , $i = 1, \dots, N$, where u is the solution of the system and $x_i = ih$.

PROBLEM 5: (10 points) Define the polynomial

$$p(t) = 1 + t + \dots + t^{n-1}$$

with coefficients all 1's. We shall try to recover these coefficients from the n values of the polynomial $p(i + 1)$, $i = 1, \dots, n$. Note that

$$p(i + 1) = \sum_{j=1}^n (1 + i)^{j-1} = \frac{1}{i} [(1 + i)^n - 1].$$

If the coefficients of the polynomial are denoted by $x_1, \dots, x_n$, we write

$$\frac{1}{i} [(1 + i)^n - 1] = p(i + 1) = \sum_{j=1}^n (i + 1)^{j-1} x_j \quad \Leftrightarrow \quad Ax = b,$$

where $A = (a_{i,j})$, $a_{i,j} = (1 + i)^{j-1}$, $b_i = \frac{1}{i} [(1 + i)^n - 1]$. Write a program to solve the resulting system using GE. Run this program for $n = 4, 5, 6, 7, 8, 9, 10, \dots, 20$. Create a table for the obtained coefficients $x_1, \dots, x_n$ (which we know are all 1's). Describe your observations. (Hint: The difficulties you observe are due to the fact that the matrix A is ill-conditioned, i.e. after a certain value of n the roundoff error is propagated and magnified and thus most of the computed values for x_i are not correct.)